• ❑ Floor plans	Room layouts for sensor placement planning	High	Medium	Generate from blueprint
• ❑ 3D mesh model	Visual representation for data contextualization	Low	High	Use photogrammetry or 3D modeling software
• ❑ Coordinate reference system	Local coordinates sufficient	Low	High	Precise global coordinates not required

Classifying each information source with its importance versus the effort it will take to incorporate into your twin helps you to prioritise what to focus on first. I will start with those sources that have a high priority and relatively low effort, before moving on to those with high priority but higher effort, before finally tackling those with lower priority.

2.7 Summary

- When building a digital twin and looking to capture physical systems for digital representation, start with objectives, not technology. Define specific, measurable business outcomes before deciding how to capture physical systems for digital representation, and at what level of fidelity.
- Historical documentation that captures designs, drawings, and the state of systems can hold critical data that can be extracted into a machine readable format using OCR and computer vision, providing the temporal context essential for predictive digital twins.
- Photographs and videos are an important data source when representing physical systems, both as representations of the physical environment at a point in time and as a source to build more complex 3D models or extract information from.
- Three main 3D representation approaches serve complementary purposes: mesh-based models provide efficient visualization for web browsers, parametric models enable precise engineering and design